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Round trip container

Abstract

A method for forming a round trip container includes stamping a container body having a receiving cavity and an opening with a proximal end of a resealable closure flap operably coupled with the container body, the resealable closure flap configured to extend over the opening. A first adhesive receiving area is provided on one of the resealable closure flap and the container body that receives adhesive to adhere the resealable closure flap to the container body when the container is placed in a closed “outbound” configuration. An adhesive portion is provided on one of the resealable closure flap and the container body with a removable covering provided over the second adhesive portion. After the container has been opened, and contents inserted into the receiving cavity, the covering is removed and the second adhesive adheres the resealable closure flap to the container body placing the container in a closed “return” configuration.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. Non-Provisional patent application Ser. No. 17/941,954 filed Sep. 9, 2022, titled “Round Trip Container” which claims the benefit of U.S. Provisional Patent Application No. 63/242,084, filed Sep. 9, 2021, the entire contents of both of which are expressly incorporated herein by reference.

TECHNICAL FIELD OF THE INVENTION

(1) The present application relates to packaging and more particularly, but not exclusively to, a round trip mailing container.

BACKGROUND

(2) Mailing containers are utilized to house various products during shipment. Many mailing containers are single-use or one-way (e.g., only designed to be sent to a single destination) and are discarded once the recipient removes the product therefrom.

(3) A round trip mailing container is designed to be sent to a user, who opens the container and places an item inside. The user then ships the round trip container back to the sender, or to another third-party.

(4) An exemplary round trip container is an outer envelope which includes a return envelope located internal to the outer envelope. As a specific example, a creditor may send an outer envelope that is addressed to a consumer. This outer envelope is an “outbound” container. The outer envelope can include a bill due to be paid as well as a return envelope addressed to the creditor or to a third party (e.g., a payment processing service). After opening the outer envelope, the customer can insert payment and/or payment information into the return envelope and place the return envelope in the mail to be delivered to the creditor. This return envelope is the “return” container.

(5) Some round trip containers are designed for carrying three-dimensional objects, which are often fragile, to either be returned by a consumer to the company or to be sent by the consumer to a third-party warranty provider. These types of round trip containers are frequently utilized for returns and warranty service with various consumer electronics, smartphones, tablets, computers, and the like. When the consumer receives the container, they will open it and insert the defective or unwanted item into the container. The consumer will then close the container, typically with packing tape, and will deliver the container to a shipping carrier to be returned to the company or delivered to a third-party warranty provider.

(6) Prior art round trip mailing containers are expensive and consumers are typically unwilling to pay an increased cost for goods as a result of packaging. Human intervention is typically required during the preparation of these round trip containers (e.g., in the form of inserting a return label and any padding material into the container). Labor costs contribute significantly to the overall expense of such round trip containers.

(7) Another drawback with known prior art round trip containers is that as the return label is placed within the interior of the container, the containers can only be made up after an order is received from a company. Therefore, ready to ship round trip containers cannot be easily held in inventory as the packaging company would not be able to insert the desired return label within the ready to ship container (e.g., as it is sealed).

(8) It is believed that the demand for round trip containers from various companies varies significantly from day to day. For example, with regard to smartphones, consumers often trade-in an older smartphone when a new desirable model becomes available. However, smartphone producers are often unable to accurately determine how many consumers will trade-in an older smartphone for a new popular model.

(9) One smartphone fulfillment house has seen the required quantity of round trip containers per day vary between around 3,000 containers per day up to around 16,000 containers per day.

(10) Therefore, further technological developments are desirable.

SUMMARY

(11) The present application is directed to round trip mailing containers, round trip mailing labels, round trip mailing systems and kits, and methods of forming round trip mailing containers and labels.

(12) One form of the present application is directed to a round trip mailing container having a container body defining a receiving cavity. An opening in the container body provides access to the

receiving cavity. A proximal end of a resealable closure flap is operably coupled with the container body, and the resealable closure flap is configured to extend over the opening.

(13) A first adhesive receiving area located at one of the closure flap and the container body. The first adhesive receiving area is configured to receive an adhesive to adhere the closure flap to the container body, when the container is placed in a first closed configuration.

(14) An adhesive portion is located on one of the closure flap and the container body, and a removable covering is disposed over the adhesive portion. In this manner, when the container is placed in the first closed configuration the adhesive portion will not adhere to the container body as the adhesive portion is covered with the removable covering. The second adhesive portion is configured to adhere the closure flap to the container body when the container is placed in a second closed configuration.

(15) The closure flap can include a proximal segment and a distal segment. The first adhesive receiving area can be located at the distal segment, and the adhesive portion can be located at the proximal segment.

(16) The container can include a release located at the closure flap between the proximal segment and the distal segment. The release can take the form of a pull-tab tear away strip. In this form, tearing of the pull-tab tear away strip separates the distal segment from the proximal segment and enables access to the receiving cavity.

(17) The container body can include a substantially rectangular-cuboid shape. The proximal end of the closure flap can be foldably coupled to the top panel. The proximal segment of the closure flap can define a first end of the container body. The container body can include a top panel, a bottom panel, a first side panel, a second side panel, and a second end panel opposite the first end.

(18) An adhesive can be located on the first adhesive receiving area to adhere the distal segment of the closure flap to the bottom panel.

(19) The container can include an inner panel extending upwardly from the bottom panel at the opening. After opening the container with the pull-tab tear away strip, the removable covering can be removed from the adhesive portion and the adhesive portion can adhere the proximal segment to the inner panel to place the container in the second closed configuration.

(20) The container can include a padded insert assembly configured for insertion into the receiving cavity. The padded insert assembly can include two horizontally extending pads which are affixed to and extend between two opposing walls, and wherein the padded insert assembly is configured for mechanized insertion into the receiving cavity.

(21) The container can include a round trip mailing label. This round trip mailing label, also referred to as a combined label (e.g., as it is a combined outbound and return label), can include a first label portion having a face surface that includes first shipment information. The combined label can include a second label portion having a second face surface configured to have second shipment information printed thereon. An adhesive can be located at the second label portion to releasably adhere the second label portion to the first label portion. The second label portion can be configured to be removed from the first label portion and adhered to the container with the adhesive located at the second label portion.

(22) A transparent pouch can retain the combined label, including the first label portion and the second label portion, to the container.

(23) Another form of the present application is directed to an apparatus that includes a first label portion having a face surface configured to have first shipment information printed thereon. This apparatus includes a second label portion having a second face surface configured to have second shipment information printed thereon.

(24) An adhesive is disposed on the second label portion, and the adhesive releasably adheres the second label portion with the first label portion to form a combined label. The second label portion is configured to be removed from the first label portion and adhered to a container.

(25) The apparatus can include a pouch configured to retain the combined label to the container.

The pouch can include a viewing window and the face surface can be oriented to be visible through the viewing window.

(26) An outer perimeter of the pouch can be adhered to the container, and the pouch can extend over to fully cover the combined label. The pouch can be configured to enable combined label to be removed therefrom absent tearing the second label portion (e.g., without tearing the second label portion). The pouch can be configured to enable the entire combined label to be removed therefrom absent tearing.

(27) A lower surface of the first label portion can include a release backing. The adhesive can be located on a second lower surface of the second label portion. The face surface and the second face surface can each face outwardly from the combined label on opposing sides thereof (e.g., a first side of the combined label includes the face surface and a second side of the combined label, opposite the first side, includes the second face surface).

(28) Another form of the present application is directed to a method for forming a round trip container. The method includes stamping a cardboard material sheet to form a unitary die cut including a top panel, a bottom panel, a first side panel, a second side panel, a closure flap having a proximal segment and a distal segment, wherein a proximal end of the closure flap is coupled with the top panel, and an end panel. The method includes providing a tear-away strip between the proximal segment and the distal segment of the closure flap. The method includes placing an adhesive on the proximal segment of the closure flap, wherein a removable covering is located atop the adhesive.

(29) Placing an adhesive on the proximal segment can include affixing a peel and stick adhesive strip to the proximal segment.

(30) The method can include adhering the first side panel to an inner side panel (thereby coupling the side panel and the bottom panel) and forming a container flat.

(31) The method can include orienting the container flat into a three-dimensional container body having an internal receiving cavity, wherein the side panel and the second side panel are approximately perpendicular relative to the top panel and the bottom panel. The method can include placing a mailing label on the container body and affixing a pouch over the mailing label to retain the mailing label to the container body.

(32) The method can include adhering the end panel to an inner end panel and adhering the distal segment to the bottom panel, thereby forming a closed outbound container.

(33) The method can include mechanically inserting an insert into the receiving cavity and mailing the outbound container to a first destination.

(34) The method can include providing a combined mailing label that has a first label portion bearing first shipment information, a second label portion bearing second shipment information, wherein the second label portion includes an adhesive portion, and wherein the second label portion is releasably adhered to the first label portion.

(35) The method can include tearing the tear-away strip and separating the proximal segment and the distal segment. The method can include removing the mailing label from the pouch and peeling the second label portion from the first label portion. The method can include pressing the adhesive portion of the second label portion to an exterior surface of the container body to adhere the second label portion to the exterior surface.

(36) The method can include removing the covering from the peel and stick covered adhesive and pressing the peel and stick adhesive on the closure flap against a first inner end panel and adhering the proximal segment of the closure flap with the first inner end panel, thereby forming a closed return container. The method can include mailing the closed return container to a second destination.

(37) Further embodiments, inventions, forms, objects, features, advantages, aspects, and benefits of the present application are otherwise set forth or become apparent from the description and drawings included herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The description herein makes reference to the accompanying drawings wherein like reference numerals refer to like parts throughout the several views, and wherein:
- (2) FIG. 1 depicts a perspective view of a resealable round trip container according to a first form of the present application;
- (3) FIG. 2 depicts a top view of the container of FIG. 1;
- (4) FIG. 2A depicts a top view of a combined mailing label according to a further form of the present application, having a first label portion having outbound shipment information thereon;
- (5) FIG. 2B depicts a bottom view of the combined mailing label of FIG. 2A and illustrates a second label portion having return shipment information thereon;
- (6) FIG. 2C depicts a side view of the combined mailing label and illustrates the second label portion being peeled from the first label portion;
- (7) FIG. 3 depicts a bottom view of the container;
- (8) FIG. 4 depicts an end view of the container;
- (9) FIG. 5 depicts a perspective side view of the container;
- (10) FIG. 5A depicts a bottom view of an exemplary die cutout from which the container of FIG. 1 may be formed;
- (11) FIG. 5B depicts a top view of the exemplary die cutout;
- (12) FIG. 6 depicts a top view of an exemplary container flat, which can be formed from the die cutout of FIG. 5A;
- (13) FIG. 7 depicts a bottom view of the container flat of FIG. 6;
- (14) FIG. 8 depicts a side perspective view of the container flat of FIG. 5 which has been formed into a three-dimensional container body;
- (15) FIG. 9 is a side view of the three-dimensional container body of FIG. 8;
- (16) FIG. 10 depicts a perspective view of the second end of the three-dimensional container body;
- (17) FIG. 11 depicts a perspective view of the first end of the three-dimensional container body, illustrating the resealable closure flap;
- (18) FIG. 12 depicts an end view of the container body of FIG. 8, placed in an exemplary packed for outbound configuration that includes a padded pouch insert;
- (19) FIG. 13 depicts an end view of the container body of FIG. 8, which has been placed in an alternative packed for outbound configuration that includes a dual padded insert;
- (20) FIG. 14 is an end view of the dual padded insert of FIG. 13;
- (21) FIG. 15 depicts an end view of the container of FIG. 13 as packed in a return configuration;
- (22) FIG. 16 depicts an end view of the container of FIG. 12 as packed in a return configuration;
- (23) FIG. 17 is a schematic view of an exemplary EFG assembly device according to a first form of the present invention;
- (24) FIG. 18 depicts a blueprint plan view of the EFG assembly device;
- (25) FIG. 19 depicts an exemplary insert conveyor and exemplary flat box feeder assembly of the EFG assembly of FIG. 18;
- (26) FIG. 20 depicts an exemplary filler assembly having an automated inserter which pushes inserts through an open end and into the three-dimensional container body;
- (27) FIG. 21 depicts a side view of the flat box feeder assembly and insert conveyor of FIG. 19;
- (28) FIG. 22 is enlarged perspective view of a forward, output end, of the flat box feeder assembly;
- (29) FIG. 23 is a side view of an exemplary folder assembly of the EFG assembly;
- (30) FIG. 24 is a side perspective view of the filler assembly of FIG. 20, depicting automated ram arms pushing the inserts through the open end of the container and into the container;
- (31) FIG. 25 depicts a perspective view of the filler assembly, depicting an insert being pushed

through a tapered chute and into the container;

(32) FIG. 25A depicts a frontal view of an exemplary gluer assembly of the EFG assembly, which folds the container panels at the open ends of the container and adhesively seals the ends to form a closed and packed container;

(33) FIG. 25B depicts a closed and packed container formed by the gluer assembly;

(34) FIG. 26 depicts an exemplary labeling line which can apply the label of FIGS. 2A and 2B;

(35) FIG. 27 is a perspective view of an exemplary staging area of the labeling line of FIG. 26;

(36) FIG. 28 depicts a perspective view of the labeling line, which illustrates an exemplary box feeder assembly, exemplary gluer bar, and exemplary controls;

(37) FIG. 29 depicts a side view of the box feeder assembly and gluer bar;

(38) FIG. 30 depicts a further perspective view of the box feeder assembly and depicts an exemplary separator;

(39) FIG. 31 depicts a further perspective view of the box feeder assembly and gluer bar;

(40) FIG. 32 depicts exemplary controls of the labeling line which are depicted as controlling an exemplary labeling assembly;

(41) FIG. 33 depicts a side view of the labeling assembly of FIG. 32;

(42) FIG. 34 depicts a further side view of the labeling assembly, illustrating the label applicator;

(43) FIG. 35 depicts a side view of an exemplary pouch applicator assembly of the labeling line of FIG. 26;

(44) FIG. 36 is an overhead perspective view of the pouch applicator assembly of FIG. 35;

(45) FIG. 37 depicts a perspective view of a fully assembled and labeled round trip container, illustrated in a closed outbound mailing configuration; and

(46) FIG. 38 depicts a perspective view of the pouch applicator assembly, which includes a backup pouch applicator assembly.

DETAILED DESCRIPTION

(47) For purposes of promoting an understanding of the principles of the invention, reference will now be made to the embodiments illustrated in the drawings and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended, any alterations and further modifications in the illustrated device, and any further applications of the principles of the invention as illustrated therein being contemplated as would normally occur to one skilled in the art to which the invention relates.

(48) As utilized herein the term “mail” and derivations thereof (e.g., “mailed”, “mailing”, “mails”, etc.) are intended to include all forms of container shipping and/or transport. Exemplary mailing service providers include the U.S. Postal Service, FedEx, UPS, and other shipping carriers, courier services, and post services, as are known.

(49) The term “container” is intended to include a variety of apparatuses in which objects are housed during mailing. Exemplary containers include envelopes, boxes, hard and soft packages and packaging, and other containers which are able to house an item to be mailed.

(50) Referring now to FIGS. 1-3 and 15, an exemplary round trip container assembly **100** will now be described. The round trip container assembly **100** includes a container **103** having a resealable closure flap **111** and a shipping label **130**. The container **103** is configured to be mailed from a first user to a first recipient and then the container **103** mailed by the first recipient to a second recipient, who may be the first user (e.g., thereby completing a round trip).

(51) Specifically, the container **103** is configured to be mailed from a first user (e.g., company, warranty provider, third-party service provider, etc.) to a first recipient (e.g., customer, end user, etc.). The container **103** includes an outbound shipping label, which can take the form of label **130**. The first user may desire various inserts or padding be inserted into a receiving cavity **1004** container **103**, prior to mailing the container **103** to the first recipient. The container **103** is closed with a resealable closure flap **111**.

(52) The first recipient opens the sealed container **103**. This can be accomplished utilizing a release

302 on the closure flap **111**. The first recipient can then insert an object **1504** (e.g., purchase to be returned, product or device to be serviced, etc.) within the receiving cavity **1004** of the container **103**. After the object **1504** is placed inside the receiving cavity **1004**, the first recipient reseals the container **103** utilizing a resealing adhesive portion **530** that is located on the closure flap **111**.

(53) The first recipient places a return label on the container **103** and mails the container to a second recipient (e.g., who may be the first user, a third-party warranty provider, a third-party returns servicer, etc.).

(54) The return label can take the form of label **130** which takes the form of a combined outbound and return label **130**. In this preferred form, the first recipient may remove the label **130** from a pouch **132**. The first recipient can then remove a return label portion **220** from the outbound label portion **200** and can stick the return label portion **220** on the container **103**.

(55) The round trip container **103** can be utilized to house a variety of objects **1504** to be mailed. Exemplary objects **1504** include, but are not limited to, electronics, household goods, clothing, automotive parts, toys, consumer goods, and the like. For example, the round trip container **103** as depicted is believed to be especially suited for the shipment (e.g., to be returned, serviced, traded-in, etc.) of small consumer electronic devices, which include but are not limited to phones, digital recorders, watches such as smart watches, tablets, computers, circuit boards, and the like. The round trip container **103** can be utilized for the return, servicing, and/or trade-in of smartphones, such as iPhones.

(56) It is contemplated that the dimensions and/or shape of the container **103**, the dimensions and/or shape of the receiving cavity **1004**, and the type and sizing of any inserts or padding may be altered (e.g., changed or modified) by a container **103** manufacturer to accommodate a specific object **1504**, or variety of objects **1504**, to be mailed therein.

(57) FIGS. 1, 2, and 3-5 depict the round trip container assembly **100** in a closed outbound mailing configuration **107**. The container assembly **100** includes a round trip container **103** and a shipping label **130**.

(58) The container **103** includes a container body **109** that extends between a first end **113** and a second end **115**. The container body **109** houses the object **1504** to be mailed. The container **103** includes a resealable closure flap **111**. Although the term “flap” is utilized with regard to closure flap **111**, the closure flap **111** does not need to be constructed of thin easily bent material; rather, the term flap is utilized with regard to the closure flap **111** as the closure flap **111** is foldably coupled with the top panel **102**.

(59) The resealable closure flap **111** enables the container **103** to be resealed after the container **103** has been opening (e.g., by a first recipient). After the container **103** has been mailed from the first user and received by the first recipient, the first recipient can open the container **103** and insert the object **1504** to be mailed. The first recipient can then reseal the container **103** utilizing the resealable closure flap **111**.

(60) Referring back to FIGS. 1-2, the container body **109** can include a substantially rectangular-cuboid shape **101**. The exterior **125** of the rectangular-cuboid container **103** is depicted as including a top panel **102**, a bottom panel **104**, a first side panel **106**, and a second side panel **108**. The resealable closure flap **111** has a first end panel **112** to cover and close the first end **113**. A second end panel **110** is located at the second end **115**.

(61) Referring now to FIGS. 1 and 3-5, the resealable closure flap **111** includes a proximal segment **124** and a distal segment **126**. The proximal segment **124** is coupled to the top panel **102**. The proximal segment **124** includes the first end panel **112** which is depicted as extending over the first end **113** and covering the first end **113** to close the container **103**. The distal segment **126** is depicted as being adhesively joined with the bottom panel **104** in the closed outbound configuration **107** of FIG. 3.

(62) A release **302** can be located at the resealable closure flap **111**. The release **302** can be located between the proximal segment **124** and the distal segment **126**. The release **302** provides for

opening of the container **103** through the separation of the proximal segment **124** from the distal segment **126** of the closure flap **111**.

(63) As is best illustrated in FIG. **3**, the release **302** can take the form of a pull-tab tear away strip **304**. As is known to a person of skill, pull-tab tear away strip **304** includes perforations **306**, a pull tab **308**, and a tearing strip (not shown) attached to the pull tab and located on a rear surface of the pull-tab tear away strip. To open the first end **113** of the container, the pull tab **308** is pulled upwardly and across the container **103**. As the pull tab **308** is pulled, the tearing strip lifts upwardly tearing the perforations **306**. This results in the removal of a strip of material across the closure flap **111** and provides for access into the container **103**. When the pull-tab tear away strip **304** is torn from the container **103**, the proximal segment **124** is separated from the distal segment **126**.

(64) However, it is also contemplated that the release **302** can take a variety of forms including a single perforated strip, a dotted line to be cut by a recipient, a tape strip over a physical separation between the proximal segment **124** and the distal segment **126** to be torn off by the recipient, or other release **302** mechanisms as are known to a person of skill. Instructions **310** to utilize the release **302** can be printed on the closure flap **111** or directly on the release **302**.

(65) The container body **109** is made of a material **105**. In one non-limiting form, the material **105** is a fluted, corrugated cardboard material **105**.

(66) Exemplary dimensions for the container body **109** will now be described. It is believed that the following dimensions are well suited for the container **103** to be utilized for mailing mobile phones, including smartphones.

(67) In this exemplary form, the top panel **102** has a length, between the first end **113** and the second end **115**, of about 34 centimeters (8.5 inches). The top panel **102** includes a width, between the first side **106** and the second side **108**, of about 18 centimeters (4.5 inches). The side panels **106**, **108** each include a length, between the first end **113** and the second end **115**, of about 17 centimeters (4.5 inches). The side panels **106**, **108** each include a width, between the top panel **102** and the bottom panel **104**, of about 6.5 centimeters (1.5 inches). The first end panel **112** and the second end panel **110** have a height, between the top panel **102** and the bottom panel **104**, of about 3.8 centimeters (1.5 inches). The first end panel **112** and the second end panel **110** have each have a width, between the side panels **106**, **108**, of approximately 18 centimeters (4.5 inches).

(68) However, it is contemplated that the container body **109** and the container **103** can be formed through a variety of construction techniques and can be formed from a variety of materials depending upon the desired characteristics of the container. The container body **109** and container **103** can include a variety of sizes and shapes, depending upon the specific object **1504** to be mailed and desired characteristics of the container **103**.

(69) A shipping label **130** is depicted as being located on the top panel **102** of the container **103**. A pouch **132** can be utilized to retain the shipping label **130** to the top panel **102**.

(70) Referring now to FIGS. **2A**, **2B**, and **2C**, an exemplary combined shipping label **130** will now be described. The combined label **130** has a first side **216**, which includes a first label portion **200**, and a second side **232**, which includes a second label portion **220**. Preferably, a face surface **202** of the first label portion **200** (e.g., the outbound label) and a second face surface **222** of the second label portion **220** (e.g., the return label) are each directed outwardly in an opposing relationship. Specifically, the face surface **202** is located on the first side **216** and the second face surface **222** is located on the second side **232**.

(71) The first label portion **200** includes the face surface **202**. This first label portion **200** can also be referred to as the outbound label. First shipment information **204** (e.g., outbound shipment information **204**) is printed on the face surface **202**. This outbound shipment information **204** can include the name and address of the first user who ships the outbound container **103** in a “from” area, the logo **208** of the first user, the name and address of the recipient in a “to” area, and shipping payment and/or postage **214**.

(72) A lower surface **254** of the first label portion **200** can include a release substrate or an easy

release backing **256**. This release backing **256** can be formed of silicone coated paper or of various polymers from which an adhesive **252** can easily release.

(73) The second label portion **220** includes a second face surface **222** upon which second shipment information **224** (e.g., return shipment information **224**) is printed. This return shipment information **224** can include the first recipient's name and address in a "from" area **226**, the second recipient's name and address in a "to" area **228**, and shipping payment and/or postage **230**.

(74) The second label portion **220** has a second lower surface **250** with an adhesive **252** located thereon. Contact between the adhesive **252** on the second label portion **220** and the easy release backing **256** on the first label portion **200** releasably adheres the second label portion **220** with the first label portion **200**, forming the combined label **130**. In this manner, the combined label **130** includes outbound information **204** on the first side **216** and return or inbound information **224** on the second side **232**.

(75) The first label portion **200** and the second label portion **220** can be manufactured of a stiff paper or cardboard material with sufficient strength to withstand the rigors of mail.

(76) The pouch **132** retains the label **130** to the container body **109**. As is illustrated in FIG. 2, the pouch **132** includes a viewing window **121** thorough which the face surface **202**, and outbound shipping information **204** located thereon, can be easily viewed. A perimeter **119** of the pouch **132** can be adhered to the top panel **102**; however, it is preferable that any adhesive utilized does not contact the face surface **202** and the second face surface **222**. The viewing window **121** does not include an adhesive. The face surface **202** and the second face surface **222** are preferably free from adhesive so that the combined label **130** can easily be removed from the pouch **132**. The pouch **132** covers and extends over the combined label **130**. An opening tab **117** is depicted on the pouch **132** which enables the first recipient to easily tear open the pouch **132** and remove the combined label **130** therefrom.

(77) Use of the combined label **130** will now be described in connection with FIGS. 2-2C. After receiving the container **103** from the first user, the first recipient tears open the pouch **132** by pulling on the opening tab **117**. The first recipient then removes the combined label **130** from the pouch **132**. The first recipient can then flip the combined label **130** so that the second face surface **222** on the second side **232** faces upwardly. The first recipient then peels the second label portion **220** from the easy release backing **256** of the first label portion **200**. The first label portion **200** can then be discarded or placed inside the container **103**.

(78) The second label portion **220** is then applied to the container body **109** and is adhered to the container body **109** via the adhesive **252** located on the second lower surface **250**. In this manner, the return shipment information **224** on the second face surface **222** faces outwardly from the container body **109** and can be easily read and/or scanned by a shipping carrier. After the object **1504** is placed inside the container body **109**, the resealable closure flap **111** can be sealed and the container **103** can be mailed to the second recipient, who can be the first user.

(79) Exemplary, non-limiting methods of the construction and life cycle of the container **103** will now be described in connection with FIGS. 5A-13. It is important to note that various stages and functions performed during the method are described as being performed by specific parties (e.g., cardboard manufacturer, converter, erector, filler, and shipper); however, it should be understood that the various functions and stages can be performed by one or more parties and some parties can perform more than one function.

(80) A cardboard manufacturer forms sheets of cardboard material **105** from sheets or webs of paper. As will be known to a person of skill, corrugated cardboard **105** comes in a variety of thicknesses and sizes and is usually designated by flute size (e.g., designed as A, B, C, D, E, or F flute size).

(81) A converter cuts the cardboard material **105** into a desired shape. The cardboard **105** can be cut with a stamping machine, utilizing a die, and is commonly referred to as die cutting. Referring now to FIGS. 5A and 5B, an exemplary die cut **502** will be described from which the container **103** can

be formed. FIG. 5A depicts a lower surface **512** of the die cut **502**, a portion of which will define a receiving cavity (e.g., an interior) of the container **103**. FIG. 5B depicts an upper surface **514** of the die cut **502**, a portion of which will define an exterior **125** of the container **103**.

(82) Preferably, the container **103** is formed of a unitary (e.g., single piece) of cardboard **105** to lessen assembly, and therefore labor, costs. The die cut **502** is a unitary structure and includes the top panel **102**, the bottom panel **104**, the first side panel **106**, the second side panel **108**, the resealable closure flap **111**, and the second end panel **110**. The die cut **502** can include an inner side panel **540**, a first inner end panel **536**, and a second inner end panel **536**. The die cut **502** can also include wing panels **534**.

(83) The die cut **502** is depicted as including fold lines **504**. The various panels included in the die cut are folded at fold lines **504** to form the container body **109**. The fold lines **504** can be stamped into the die cut **502** for ease of folding, as is known.

(84) The resealable closure flap **111** is part of the unitary die cut **502**. A proximal end **544** of the closure flap **111** is depicted as being connected to the top panel **102** and a distal end **546** of the closure flap **111** extends outwardly from the die cut **502**.

(85) The release **302**, which is depicted as pull-tab tear away strip **304**, is located between the proximal segment **124** of the closure flap **111** and the distal segment **126** of the closure flap **111**. When utilized to open the container **103**, the pull-tab tear away strip **304** will tear the closure flap **111** along axis **510**, thereby separating the proximal segment **124** from the distal segment **126** and enabling access to the interior of the container **103**. A reverse side **508** of the pull-tab tear away strip **304** is shown in FIG. 5A.

(86) The release **302** can be integrally formed into the cardboard material **105** and cut therefrom with the die cut **502**; however, depending upon the release **302** selected and the desired process utilized to form the container, the release **302** can also be integrated into the closure flap **111** later in the process (e.g., such as when the second adhesive portion **530** is applied).

(87) An adhesive receiving area **520** is located on the distal segment **126**. An adhesive receiving area **522** is depicted on the side panel **106** and an adhesive receiving area **524** is depicted on the second end panel **110**.

(88) FIG. 5B depicts the die cut **502** as including second adhesive portion **530**. The second adhesive portion **530** has a removable covering **532** located thereon. The second adhesive portion **530** is not utilized to sealingly close the first end **113** until the container **103** has been received and opened by the first recipient. Therefore, the removable covering **532** prevents the second adhesive portion **530** from prematurely sticking to the container **103** or to any inserts placed therein on the outbound trip to the first recipient. The second adhesive portion **530** can take the form of a peel and stick adhesive which includes a backing suitable for adherence to the proximal segment **124**.

(89) The second adhesive portion **530** was included in FIG. 5B to clearly depict the location of the second adhesive portion **530** on the container **103**; however, it should be understood that the second adhesive portion **530** can be attached later in the process.

(90) It is also contemplated that the container **103** can be configured to reseal for mailings beyond a round trip (e.g., single outbound and single return). For example, a resealable closure flap **111** can be included on the second end **115** and the first end **113**. The proximal portion **124** can include two or more alternating second adhesive portions **530** and releases **302**.

(91) To form a container flat **602**, as is depicted in FIG. 6, the converter can also fold the die cut **502** at the fold line **504** between the second side panel **108** and the bottom panel **104** and can fold the die cut **502** between the first side panel **106** and the top panel **102**. The converter may apply an adhesive on the adhesive receiving area **522** and adhere the first side panel **106** to the inner side panel **540**. FIGS. 6 and 7 depict an exemplary container flat **602**.

(92) As utilized herein, the term “adhesive” is intended to include a wide variety of adhering substances, which include but are not limited to glues, epoxies, various polymer based adhesives and adhesive strips, and other sticky substances and adhesives as would be known to a person of

skill.

(93) The container flat **602** is not yet formed into container **103**. However, as the container flat **602** has not yet been fully constructed (e.g., is flattened) the container flat **602** requires much less shipping space than the fully constructed container **103**. This container flat **602** is then shipped to and erector. The erector will be described herein as serving the roles of an erector, filler, and shipper as it is believed that the integration of these operations frequently occurs.

(94) The erector receives the container flat **602** and forms the container flat **602** into a three-dimensional container body **802**, shown in FIGS. **8-11**. One primary difference between the container **103** in the closed outbound configuration **107** and the container **103** formed into the three-dimensional container body **802**, is that in the three-dimensional container body **802** the container **103** has the first end **113** in an open configuration and the second end **115** in an open configuration.

(95) The erector orients the container flat **602** into this three-dimensional container body **802** that has an internal receiving cavity **1004**. To achieve this, the erector orients the first side panel **106** and the second side panel **108** to be approximately perpendicular relative to the top panel **102** and the bottom panel **104** (e.g., the first side panel **106** and the second side panel **108** are placed in a substantially parallel orientation, and the first side panel **106** and the second side panel **108** intersect the top panel **102** and the bottom panel **104** at an angle of around 90 degrees.

(96) Serving in the role of a filler, the erector places one or more inserts **1202** through the opening **1002** in the second end **115** and/or the opening in the first end **113**.

(97) Referring now to FIGS. **10-13**, the insert **1202** can include various padding members, such as a padded pouch **1206** (e.g., which can be a padded envelope as are known), a dual padded insert **1302**, or other padding. The insert **1202** can also include instructions **1204** for the first recipient to trade-in or return the object **1504**, and/or various other literature such as advertising materials.

(98) Preferably, the insert **1202** is constructed in a manner to permit the insert **1202** to be placed in the receiving cavity **1004** of the container **103** mechanically. For example, the insert **1202** can travel on a first conveyor belt and the three-dimensional container body **802** can travel on a second conveyor belt next to the first conveyor belt. The three-dimensional container body **802** can be oriented on the second conveyor belt such that the open first end **113** or the open second end **115** is oriented toward the insert **1202** on the first conveyor belt. A machine, such as a robotic arm, pneumatic cylinder, or various other electro-mechanical devices, can push the insert **1202** from the first conveyor belt through the first end **113** or second end **115** and into the receiving cavity **1004**.

(99) Referring briefly to FIGS. **13-14**, an exemplary dual padded insert **1302** will now be described. This dual padded insert **1302** includes a first padding member **1402** and a second padding member **1403**. Opposing walls **1406**, **1408** extend upwardly from a base member **1404**. The base member **1404** and walls **1406**, **1408** can be constructed of a cardboard or cardstock-type material **1410**.

(100) The first padding member **1402** and the second padding member **1403** extend between the walls **1406**, **1408** and have a substantially parallel relationship. A space **1402** can be provided between the padding members **1402**, **1403**. Opposing ends of the first padding member **1402** and the second padding member **1403** can be adhered and/or mechanically fastened to the walls **1406**, **1408**. A pocket **1304** is located between the padding members **1402**, **1403** or the padding members **1402**, **1403** can compress to provide for the pocket **1304**. The pocket **1304** is configured to receive the object **1504** from the first recipient.

(101) The padding members **1402**, **1403** are structured to provide cushion to the object **1504** when the container **103** is mailed in a return configuration. Such cushion can be very important when the object **1504** is easily breakable (e.g., as is the case with smartphones and many other small electronic devices).

(102) The padding members **1402**, **1403** are depicted as having a pleated **1412** accordion-type shape. However, it is contemplated that various padding members **1402**, **1403** can be utilized in the

dual padded insert **1302** including foams, inflated plastics such as bubble wrap, rubber padding materials, and other cushioning members as are known.

(103) Advantageously, this dual padded insert **1302** can be readily mechanically inserted into the receiving cavity **1004**. This ease of mechanical insertion is believed to result from the base member **1404** providing for ease of sliding, and the interaction between the walls **1406**, **1408** and padding members **1402**, **1403** maintains the overall dimensions of the insert **1302** (e.g., so that the insert **1302** will not compress and expand as the machine begins to push the insert **1302**).

(104) When the inserts **1202** have been placed into the receiving cavity **1004**, the container **103** is now in a packed for outbound configuration **1208**, as shown in FIGS. **12** and **13**. To achieve the closed outbound configuration **107**, the ends **113**, **115** of the container **103** should be closed and the label **130** should be applied.

(105) Referring back to FIGS. **1-2** and **5A-5B**, the erector then closes the first end **113** and the second end **115** to close the container **103**. To close the first end **113**, adhesive is applied to the adhesive receiving area **520** located on the distal segment **126** of the closure flap **111**. It is important to note that the adhesive receiving area **520** is distinct from the second adhesive portion **530**, and the second adhesive portion **530** is not utilized to achieve the closed outbound configuration **107**.

(106) The erector then folds the wing panels **534** at the first end **113** and the first inner flap panel **538** toward the opening **902**. The closure flap **111** is folded at fold lines **504** so that the proximal end of the closure flap **111** covers the opening **902** to serve as the first end panel **112**. The distal segment **126** of the closure flap **111**, and the adhesive thereon, is pressed against the bottom panel **104** such that the adhesive on the adhesive receiving area **520** adheres the distal segment **126** to the bottom panel **104**.

(107) To close the second end **115**, adhesive is placed on the adhesive receiving area **524** of the second end panel **110**. The erector folds the wing panels **534** at the second end **115** and the second inner end panel **536** toward the opening **1002**. The second end panel **110** is then pressed against the second inner end panel **536** so that the adhesive on the adhesive receiving area **524** adheres the second end panel **110** to the second inner end panel **536** yielding a closed packed configuration, as will later be described as closed and packed container **2606** in FIG. **27**.

(108) Although specific adhesive receiving areas **520**, **522**, **524** have been described, adhesives can be applied to the opposing surface with equal effect (e.g., rather than placing adhesive on the second end panel **110** to adhere the second end panel **110** to the second inner end panel **536**, adhesive can be applied to the second inner end panel **536** to adhere the second end panel **110** thereto).

(109) The erector can then serve in the role of a shipper and can print the outbound shipping information **204** and return shipment information **224** onto the label **130**, which can be a combined label **130**. The combined label **130** includes a first side **216** having a first face **202** which includes outbound shipping information **204** and a second side **232** which includes a second face surface **222** that has return shipment information **224** located thereon.

(110) As will be understood, the specific outbound shipping information **204** and return shipment information **224** will depends upon the person and/or entity who is the first user, the person and/or entity who will be the first recipient, and the person and/or entity who will be the second recipient. As a non-limiting example, in the case of a container **103** configured for a smartphone, the user who sends the container **103** in a closed outbound configuration **107** can be the phone carrier (e.g., Sprint, T-Mobile, Verizon, AT&T, etc.) the manufacturer (e.g., Apple, Samsung, Google, etc.); however, the user can also be a third-party distributor, servicer, or logistics fulfillment company who handles the returns and/or servicing for the manufacturer or carrier.

(111) The first recipient that receives the container **103** in the closed outbound configuration **107** will likely be a consumer or end user who desires to return, trade-in, or have service work performed on the smartphone. The first recipient mails the container **103** in a return mail

configuration.

(112) The second recipient of the container **103**, who receives the container in return configuration can be the first user or a third-party that handles such inbound returned objects **1504** for the manufacturer, distributor, or carrier.

(113) After the label **130** is printed, the erector who is serving as a shipper, can place the label **130** on the container **103** with the outbound shipment information **204** directed outwardly from the container. The label **130** can be held to the container **103** with one or two small drops of adhesive to prevent the label **130** from sliding or blowing off the container **103** as the container **103** moves along a conveyor; however, the amount and type of adhesive used should be such that the label **130** can be easily removed from the container **103** without tearing.

(114) A perimeter **119** of the pouch **132** is then adhered around the label **130**. The container is now in the closed outbound configuration **107** as has been described herein. The erector can then deposit the container in the mail.

(115) When the first recipient receives the container **103** the first recipient will open the container **103**. This can be accomplished by tearing the pull-tab tear away strip **304**. Removal of the pull-tab tear away strip **304**, as has been described herein, results in a separation of the proximal segment **124** of the closure flap **111** from the distal segment **126**. The distal segment **126** will remain adhered to the bottom panel **104**. As the proximal end **544** of the closure flap **111** is foldably connected to the top panel, the proximal segment **124** can move away from the distal segment **126** adhered on the bottom panel **104**.

(116) The first recipient can gain access to the receiving cavity **1004** by folding the proximal segment **124** (e.g., being first end panel **112**), the first inner end panel **538**, and the wing panels **534** outwardly and away from the opening **902**. With the panels **112**, **538**, and **534** moved away from the opening **902**, first recipient may then access the receiving cavity **1004** and remove any instructions **1204** therefrom.

(117) The first recipient can then place the object **1504** to be mailed, presently taking the form of smartphone **1504** in this method, to be mailed in the receiving cavity **1004**. Depending upon the instructions **1204** and any inserts **1202** which are provided in the outbound container **107**, the first recipient may place the smartphone **1504** directly into the receiving cavity **1004**, within a padded pouch **1206** to be located within the receiving cavity **1004** (as shown in FIG. 16), into the pocket **1304** of a dual padded insert **1302** between the first padding member **1402** and the second padding member **1403** (as shown in FIG. 17), or with other padding or inserts which are provided.

(118) The first recipient can open the pouch **132**, by pulling opening tab **117**. The combined label **130** can be pulled from the pouch, separating from any small glue droplets which adhered the combined label **130** to the container **103** during manufacture. The first recipient can now peel the second label portion **220** from the first label portion **200** and can adhere the second label portion **220** to the exterior **125** of the container **103**. The container **103** is now in a packed for return configuration **1502**. The first end **113** of the container **103** can be closed and the container **103** shipped.

(119) The first end **113** of the container **103** is sealed closed utilizing the second adhesive portion **530** on the proximal segment **124** of the resealable closure flap **111**. The first recipient can fold wing panels **534** and the first inner end panel **538** inwardly toward the opening **902**. The first recipient can remove the covering **532** from the second adhesive portion **530** and can press the proximal segment **124** toward and against the inner end panel **538**. With the proximal segment **124** pressed against the inner end panel **538**, the adhesive of the second adhesive portion **530** will adhere the proximal segment **124** to the inner end panel **538** closing and sealing the first end **113**.

(120) With the second label portion **220** adhered to the container **103**, the smartphone **1504** inserted into the receiving cavity **1004**, and with the first end **113** adhered closed, the container **103** is in a ready for return mail configuration. The first recipient can now place the container **103** in the mail to be delivered to the second recipient.

(121) The second recipient receives the container **103**, opens the container **103**, and removes the object **1504**. The container **103** can be discarded.

(122) An exemplary method of manufacturing the container **103**, and machinery utilized to construct the container **103** will now be described. The container **103** can be constructed through a process which is highly automated and mechanized, reducing labor costs. One benefit of the round trip container **103** is that it is capable of being inventoried in as a closed and packed container **2606** (e.g., depicted in FIGS. **25A** and **27**, in which inserts **1202** have been placed in the receiving cavity **1004** and where the ends **113**, **115** of the container **103** are closed and sealed; however, the label **130** has not been applied). Prior art round trip containers could not be inventoried in a closed and packed configuration as the return label would need to be placed within the container prior to closing and sealing the container.

(123) In this manner, the container **103** when inventoried as a closed and packed container **2606** can quickly be transitioned to the fully closed outbound configuration **107** by printing and applying the combined label **130** and the pouch **132** to the exterior of the container **125**. The ability to readily produce on demand round trip containers **103** in a fully closed outbound configuration **107** is believed to be highly advantageous as the demand for such can vary widely from day to day and week to week.

(124) For example, the introduction of a highly desirable new product, such as a new Apple iPhone release, will typically result in an initial surge of sales. This initial surge of sales can bring a surge of phones to be returned (e.g., as part of a trade-in program). Additionally, carriers such as Verizon, T-Mobile, AT&T, etc. can run deals on products, which can result in an increased demand for round trip containers **103** through trade-in offers or phone buy-out returns. For example, some carrier offers have provided a deeply discounted smartphone in return for extending the customer's contract with the carrier or switching the customer's business from one carrier to another. Demand for containers **103** can also surge due to manufacturer recalls or consumers becoming aware of a given issue with a smartphone (e.g., battery life issues).

(125) An exemplary process performed by an erector/filler/gluer assembly ("EFG Assembly") will now be described. FIG. **17** is a schematic representation of the steps that occur during the process performed by the EFG Assembly. It is contemplated that some steps of the process **1700** are manually performed by humans; however, most steps within process **1700** are preferably automated and mechanized.

(126) Referring now to FIGS. **17-18**, the step **1702** depicts that the container flats **602** are placed staging area **1702**. The container flats **602** are conveyed **1704** to a flat box feeder assembly **1804**. At step **1706** the container flat **602** is folded by a folder assembly **1804** to form a three-dimensional container body **802** having a substantially rectangular-cuboid shape **101**. The three-dimensional container body **802** is conveyed **1708** to a filler assembly **1808**. The ends **113** and **115** are depicted in an open configuration on the three-dimensional container body **802**.

(127) At step **1710**, the filler assembly **1808** places any inserts **1202**, such as dual padded insert **1302**, padded pouch **1206**, and/or instructions **1204** through an open end **113** or end **115** and into the receiving cavity **1004** of the container **103**.

(128) The container **103**, with the insert **1202** located therein, is then conveyed **1712** to a gluer assembly **1810**.

(129) At step **1714**, the gluer assembly **1810** folds the wing panels **534** at the first end **113** and the first inner flap panel **538** toward the opening **902**. The wing panels **534** at the second end **115** and the second inner end panel **536** are folded toward the opening **1002**, which can be performed by the gluer assembly **1810** at the same time as the folding of the panels of the first end **113**. The closure flap **111** is folded.

(130) The gluer assembly **1810** adheres the distal segment **126** of the closure flap **111** to the bottom panel **104**. The second end panel **110** is adhered by the gluer assembly **1810** with the second inner end panel **536**. A closed and packed container **2606** has now been formed.

(131) The closed and packed container **2606**, which is a semi-finished mailing container **103** that merely fails to include label **130** on an exterior thereof, is then conveyed **1716** to a box collection area. At step **1718**, the closed and packed containers **2606** can be either manually or automatically stacked. The closed and packed containers **2606** can remain in inventory, be forwarded to a warehouse, or sent directly to a labeling line **2600** as will be described in connection with FIGS. **26-38**.

(132) The conveying and/or conveyors **1704**, **1708**, **1712**, and **1716** can be performed by a singular conveyor belt or multiple conveyor belt assemblies configured to move the container **103**, at various stages of construction, through the EFG assembly **1800**.

(133) Referring now to FIGS. **18-25B**, the machinery of the EFG Assembly **1800** will now be described. This EFG Assembly **1800**, also known as a cartoner, includes a flat box feeder assembly **1802**, a folder assembly, a folder assembly **1808**, a gluer assembly **1810**, an insert conveyor **1806**, and a box conveyor **1811** which is located next to and parallel to the insert conveyor **1806**.

(134) A stack of container flats **602**, depicted as array **2101**, is provided to the flat box feeder assembly **1802**. As shown in FIGS. **21** and **22**, the flat box feeder assembly **1802** includes a rear pusher **2104**, opposing aligner bars **2103**, forward stops **2105**, a feed mechanism **2107**, and a box manipulator **2102**.

(135) The rear pusher **2104** presses against the array **2102** to enable container flats **602** to be continuously and consistently provided to the feed mechanism **2107**. As the container flats **602** are pushed toward the feed mechanism **2107**, the container flats **602** are oriented on their side, as the array **2101**, between the opposing aligner bars **2103**. The forwardmost container flat **602** (e.g., the container flat **602** to be acted on by the feed mechanism **2107** next) contacts forward stops **2105**.

(136) The feed mechanism **2107**, which is depicted as a mechanized push feed **2107**, serves to feed individual container flats **602** (e.g., one at a time) to the box manipulator **2102** and prevents multiple container flats **602** from being simultaneously released to the box manipulator **2102**. The box manipulator **2102** can take the form of rotatable suction cups **2102**, which utilize a vacuum to grab the container flat **602**. The rotatable suction cups **2102** then rotate the container flat **602** to be placed on the box conveyor **1811** in the desired orientation.

(137) A folder assembly **1804** then folds the container flats **602** to form a three-dimensional container body **802** having a substantially rectangular-cuboid shape **101**. Referring now to FIG. **23**, moving stops **2304** can interact with the first side panel **106** and suction cups **2102** can interact with the bottom panel **104** (depicted as located upwardly) to orient the side panels **106**, **108** vertically (e.g., with the side panels **106**, **108** perpendicular to the top panel **102** and bottom panel **104**).

(138) Alternatively, the moving stops **2304**, which include a vertically extending member **2309**, can interact with the first side panel **106** and opposing fixed stops (not shown) can interact with the second side panel **108**. Pressing the vertically extending member **2309** against side panel **106** and pressing an opposing vertically extending member of the opposing fixed stops (not shown) against the second side panel **108** will also form the container flat **602** into the three-dimensional container body **802**.

(139) As has been previously discussed, the ends **113** and **115** are depicted in an open configuration on the three-dimensional container body **802** formed by the folder assembly **1804**. The three-dimensional container body **802** is conveyed by the box conveyor **1811** to a filler assembly **1808**.

(140) Referring now to FIGS. **18-20**, an insert conveyor **1806** is located next to and parallel with the box conveyor **1811**. The box conveyor **1811** is depicted as being a chain drive conveyor **2006**; however, it is contemplated that the conveyors described herein, including box conveyor **1811** and insert conveyor **1806**, can include a number of conveyor belt types and drive types (e.g., belt, track, chain and pusher, and the like) as are known.

(141) The insert conveyor **1806** includes a plurality of insert holding trays **1904** which are separated by spacers **1906**. Inserts **1202** are placed in the insert holding trays **1904**. An automatic

insert feeder system (not shown) can be provided to feed the inserts **1202** into the holding trays **1904** without human intervention. As has been described, the inserts can take the form of dual padded insert **1302**, padded pouch **1206**, instructions **1204**, other padding members, and/or literature including advertisements.

(142) The insert conveyor **1806** transports the inserts **1202** toward the filler assembly **1808**. The filler assembly **1808** transfers the inserts **1202** from the insert conveyor **1806** into the three-dimensional container bodies **802** on the box conveyor **1811**. The filler assembly **1808** is constructed to operate absent human intervention.

(143) Referring now to FIGS. **20**, and **24-25** the filler assembly **1808** converts the three-dimensional container body **802** into a packed for outbound configuration **1208**. The filler assembly **1808** includes an automated inserter **2002**.

(144) Tapered chutes **2004** can be located between the insert conveyor **1806** and the box conveyor **1811**. The tapered chutes **2004** are depicted as tapering inwardly toward the box conveyor **1811**. When the insert **1202** is pushed through a tapered chute **2004**, the tapered chute **2004** correctly aligns the insert **1202** with the opening **1002** of the open second end **115** of the container **103**. It is believed that chutes **2004** will reduce the likelihood that the insert **1202** will catch on an edge of the open second end **115** and reduces the likelihood that the insert **1202** will compress and extend outwardly when pushed by the automated inserter **2002**.

(145) The insert conveyor **1806** transports the inserts **1202** toward the chutes **2004**. The box conveyor **1811** transports the three-dimensional container bodies **802** toward the exit end of the chutes **2004**. When the open second end **115** of a three-dimensional container body **802** and an insert holding tray **1904** are both correctly aligned with a chute **2004**, which can be sensed via electronic eyes (not shown), the automated inserter **2002** can push the insert **1202** through the opening **1002** at the second end **115** and into the receiving cavity **1004** of the container **103**. The automated inserter **2002** can utilize an arm, piston, actuator, or other pushing mechanism to push the insert **1202**. The container **103** is now in a packed for outbound configuration **1208**.

(146) Referring now to FIGS. **18** and **25A-25B**, a gluer assembly **1810** closes and seals the first end **113** and the second end **115** to form a closed and packed container **2606**. The gluer assembly **1810** folds the wing panels **534** at the first end **113**, the closure flap **111**, and the first inner flap panel **538** toward the opening **902**. The wing panels **534** at the second end **115** and the second inner end panel **536** are folded toward the opening **1002**, which can be performed by the gluer assembly **1810** simultaneously along with the folding of the panels **538**, **534** of the first end **113**.

(147) The gluer assembly **1810** includes adhesive applicators **2402** which apply adhesive to the adhesive receiving areas, as have been previously described. The gluer assembly adheres the distal segment **126** of the closure flap **111** to the bottom panel **104**. The second end panel **110** is adhered by the gluer assembly **1810** with the second inner end panel **536**. A closed and packed container **2606** has now been formed.

(148) The closed and packed container **2606**, which is a semi-finished mailing container **103** that does not yet include label **130** on an exterior thereof, is then conveyed **1716** to a box collection area. The closed and packed containers **2606** can be placed in warehouse inventory to provide containers **2606** during surge periods; however, the containers **2606** can be labeled directly from the EGF assembly **1800**.

(149) Referring now to FIGS. **26-38**, an exemplary labeling line assembly **2600** will now be described. The containers **2606** can be moved from inventory and placed on the labeling line assembly **2600** to quickly have the labels **130** and pouches **132** adhered thereto.

(150) As will be appreciated to a person of skill, having almost fully completed containers **2606** (e.g., merely lacking label **130** and pouch **132** relative the fully closed outbound configuration **107**) provides for a greater output of containers **103** in a fully closed outbound configuration **107** at times of high demand, enabling more containers **103** to be shipped to first recipients over a given period of time. It is also important to note that the present processes described herein remove the

need for the containers **103** to be hand-filled (e.g., especially with regard to inserts **1202**).

(151) The prior art process of placing a return label by hand into the interior of the box, then sealing the box, is no longer required. As will be appreciated, the overall output of containers **103** in a fully closed outbound configuration **107** has been enhanced significantly. Additionally, the overall costs of container **103** production are reduced due to the decrease of labor costs.

(152) Referring back to FIG. **26**, an exemplary labeling line assembly **2600** includes a container input conveyor **2604**, a box feeder assembly **2620**, a gluer bar **2612**, a label applying assembly **2602** (which includes a label feeder assembly **2610** and a label applicator **2614**), a line conveyor **2618**, and a pouch applicator assembly **2616**.

(153) Referring now to FIGS. **27** and **28**, semi-finished containers **2606** are stacked in a holding area **2618**. The containers **2606** are packed, closed, and sealed and awaiting application of a label **130** and pouch **132**.

(154) A container input conveyor **2604**, which includes a conveyor belt **2608**, moves the containers **2606** from the holding area **2618** toward a box feeder assembly **2620**. The box feeder assembly **2620** feeds one container **2606** at a time onto the line conveyor **2618**.

(155) Referring now to FIGS. **29-31**, the container input conveyor **2605** moves the containers **2606**, in an incremental batch fashion, toward the box feeder assembly **2620**. The box feeder assembly **2620** is depicted as including feeding walls **2621** which retain the containers **2606** in an aligned stack. The feeding walls **2612** of the box feeder assembly **2620** are depicted as being a three-walled, substantially rectangular open trough for receiving the stacked containers **2606**. The box feeder assembly **2620** then deposits the containers **2606**, one at a time, onto the line conveyor **2618**.

(156) The box feeder assembly **2620** can include a separator **2628** and a pressure application wheel **2630**, as is best shown in FIG. **30**. The separator **2628** is designed to provide an adequate spacing between the containers **2606** as they are deposited on the line conveyor **2618**.

(157) The pressure application wheel **2630** takes the form of an arm having a wheel on a distal end thereof. The pressure application wheel **2630** applies pressure to the container **2606** to hold the container **2606** against the conveyor **1618**. The pressure application wheel **2630** additionally ensures that only a single container **2606** is fed by the box feeder assembly **2620** at a time (e.g., ensuring that a stack of containers **2606** is not fed, which would result in a jam).

(158) FIG. **30** also depicts exemplary controls **2622** for the box feeder assembly **2620**, exemplary controls **2624** for the gluer bar **2612**, and exemplary controls **2625** for the label applicator assembly **2602**.

(159) A glue applicator **2612**, depicted as a gluer bar **2612**, is located downstream of the box feeder assembly **2620** on the line conveyor **2618**. This gluer bar **2612** is configured to place a droplet or two of glue on the top panel **102** of the container **2606**, so as to hold the label **130** in place as the container **2606** moves down the conveyor **2618**. These droplets of glue firmly hold the label **130** in place and prevent various environmental factors of the conveyor **2618** (e.g., wind from moving down the conveyor, shifting movements of the conveyor, etc.) from causing the label **130** to move on the container **2606** or to fall from the container **2606**.

(160) As has been described herein, the combined label **130** is removed by the first recipient and a second label portion **220** is employed by the first recipient as a return label (e.g., the second label portion **220** is peeled from the first label portion **200** and the second label portion **220** is adhered to the exterior **125** of the container **103**). Therefore, the glue composition and amount of glue emitted by the gluer bar **2612** should not be such that the first recipient will tear or destroy the label **130** when attempting to remove the label from the top panel **102** of the container **103**. Rather, the glue emitted by the gluer bar **2612** should have a gentle hold (e.g., which can be similar to the adhering strength found on a Post-It Note®, which is capable of adhering to a surface but can later be removed without tearing and/or removing the print from the surface when removed).

(161) Referring now to FIGS. **32-34**, the label applicator assembly **2602** will now be described. The label applicator assembly **2602** is located downstream of the gluer bar **2612** on the line

conveyor **2618**. The label applicator assembly **2602**, which is best shown in FIG. 33, includes a label supply stacker **2632**, a label feeder **2610**, and a label applicator **2614**.

(162) Stacked labels **130** are placed into the label supply stacker **2632**. The label feeder **2610** receives the labels from the supply stacker **2632** and provides them, in a singular fashion, to the label applicator **2614**. The label feeder **2610** is depicted as a Straight Shooter® brand label feeder **2610**; however, it is contemplated that a variety of label feeders **2610** can be utilized to feed the labels **130** to the label applicator **2614**.

(163) The label feeder **2610** moves the labels **130** from the stack at the supply stacker **2632** and positions the label such that the label applicator **2614** can apply the label to the top panel **102** of the container **2606**. The label applicator **2614** can include a pressing mechanism, shown as pressing wheel **3400**, which serves to press the label **130** firmly against the glue drops which the gluer bar **2612** deposited on the top panel of the container **2606**.

(164) The label applicator **2614** is oriented relative to the containers **2606** on the conveyor **2618** so that the labels **130** are applied to the top panels **102** of the containers **2606** in a substantially uniform and even manner. The uniformity of label **130** application by the label applicator **2614** is especially important as a pouch **132** will be applied over the label, as will be described hereinafter.

(165) Referring now to FIGS. 35-38, an exemplary pouch applicator assembly **2616** will be described. The pouch applicator assembly **2616** is located downstream of the label applicator **2614**. The pouch applicator assembly **2616** is depicted as including a pouch **132** supply roll **2636**, a substrate take up reel **2650**, a number of rollers **2640**, **2642**, and **2644**, and a pouch applicator **2648**. The pouch applicator assembly **2616** is configured to apply a pouch **132** over the label **130** which firmly retains the label **130** to the top panel **102**.

(166) As has been briefly described, the pouches **132** are not entirely covered with. Rather, the adhesive is only applied to a perimeter **119** of the pouch **132**. The viewing window **121**, which is depicted as fully covering the label **130**, is adhesive-free and transparent. Therefore, the label **130** can be easily seen through the viewing window **121** and the label **130** will not stick to the pouch **132** when the first recipient opens the pouch **132** via tab **117** to remove the label **130** therefrom.

(167) The pouches **132** are applied to an easy release substrate **2652**. The pouches **132** and substrate **2652** can be viewed as taking a peel and stick form (e.g., the pouches **132** are peeled from the substrate **2652** to be adhered over the label **130**). The pouches **132** can be provided on the substrate **2652** in the form of a pouch **132** supply roll **2636**. As the substrate **2652** is pulled toward the pouch applicator **2648**, the pouch **132** supply roll **2636** rotates around the spool roller **2638**. The pouches **132** and substrate **2652** then traverse a series of rollers **2642**, **2640**, and **2644** prior to arrival at the pouch applicator **2648**. These rollers **2642**, **2640**, and **2644** ensure that the substrate **2652** is maintained in tension, and one or more of the rollers **2642**, **2640**, and **2644** may be drive rollers (e.g., providing a motive force to the substrate **2652**).

(168) The pouch applicator **2648** is located just above the containers **2606** on the line conveyor **2618**. The pouch applicator **2648** peels a pouch **132** from the substrate **2652** and the pouch applicator **2648** presses the adhesive on the perimeter **119** of the pouch **132** on the top panel **102** around the label **130**. A properly positioned pouch **132** is located over the label **130** so that the label **130** is placed within the confines of viewing window **121** of the pouch **132** and is visible therethrough. The adhesive on the outer perimeter **119** of the pouch **132** preferably does not contact the label **130**.

(169) Once the pouch **132** has been applied over the label **130** onto the top panel **102**, the now empty substrate **2652** (e.g., as the pouch **132** was peeled therefrom) is spooled around a substrate take up reel **2650**.

(170) A controller **2634** is utilized to control the pouch applicator assembly **2616**. A back-up pouch applicator assembly **2654** is provided. The pouch applicator assembly **2616** and the back-up pouch applicator assembly **2654** operate in a similar fashion to a primary and back-up projector in a movie theater. Specifically, the pouch applicator assembly **2616** will feed and apply pouches **132**

on containers **2606** until its supply reel **2636** runs out of pouches **132** and is empty. At that point, the back-up pouch applicator assembly **2654** takes over, continuing to apply pouches **132** over the labels **130** on the containers **2606** until its supply reel **2680** is also exhausted.

(171) When the back-up pouch applicator assembly **2654** is operating, the supply reel **2636** of the pouch applicator assembly **2616** can be replaced. This replacement preferably occurs prior to the supply reel **2680** of the back-up pouch applicator assembly **2654** running out of pouches **132**. In this manner, the pouch applicator assembly **2616** and the back-up pouch applicator assembly **2654** allow for the application of pouches **132** when the respective supply reels **2636** and **2680** are replaced, enabling continued run time of the labeling line assembly **2600**. This is believed to be highly advantageous as the labeling line assembly **2600** is highly automated; therefore, humans may not frequently be present at the labeling line assembly **2600** to provide for frequent supply reel **2636** and **2680** changes.

(172) After the pouch **132** has been applied over the label **130** with the perimeter **119** adhered to the top panel **102**, the container **103** is in the fully closed outbound mailing configuration **107**, as is depicted in FIGS. **1-2**. This finished container **103** is ready to be mailed from the first user to the first recipient as has been previously described.

(173) While the invention has been described in connection with what is presently considered to be the most practical and preferred embodiment, it is to be understood that the invention is not to be limited to the disclosed embodiment(s), but on the contrary, is intended to cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as permitted under the law.

(174) Furthermore, it should be understood that while the use of the word preferable, preferably, or preferred in the description above indicates that feature so described may be more desirable, it nonetheless may not be necessary and any embodiment lacking the same may be contemplated as within the scope of the invention, that scope being defined by the claims that follow. In reading the claims it is intended that when words such as “a,” “an,” “at least one” and “at least a portion” are used, there is no intention to limit the claim to only one item unless specifically stated to the contrary in the claim. Further, when the language “at least a portion” and/or “a portion” is used the item may include a portion and/or the entire item unless specifically stated to the contrary.

Claims

1. A method for forming a round trip container, comprising: stamping a cardboard material sheet to form a unitary die cut body, wherein the unitary die cut body includes a top panel, a bottom panel opposite the top panel with the bottom panel having a first bottom panel end and a second bottom panel end, a first side panel, a second side panel, a resealable closure flap having a proximal resealable closure flap segment and a distal resealable closure flap segment, and an end panel, wherein a proximal end of the resealable closure flap is coupled to the top panel, wherein the closure flap and the end panel are positioned on opposing sides of the top panel and are foldable along parallel axes, whereby when the unitary die cut body is folded, the folded unitary die cut body defines a unitary die cut body container having a receiving cavity with an opening providing access to the receiving cavity, and wherein the resealable closure flap is configured to extend over the opening with a proximal end of the resealable closure flap operably coupled with the first end of the top panel of the unitary die cut body container; providing a tear-away strip between the proximal resealable closure flap segment and the distal resealable closure flap segment; placing a first adhesive receiving area at one of the resealable closure flap and the second end of the bottom panel of the unitary die cut body container that is opposite the first end of the unitary die cut body container, wherein the first adhesive receiving area is configured to receive adhesive to adhere the resealable closure flap to the unitary die cut body container when the unitary die cut body container

is placed in a first closed configuration; and placing an adhesive portion located on one of the resealable closure flap and the second end of the unitary die cut body container, wherein a removable covering is disposed over the adhesive portion when the unitary die cut body container is placed in the first closed configuration, and wherein the adhesive portion is configured to adhere the closure flap to the unitary die cut body container when the unitary die cut body container is placed in a second closed configuration; wherein stamping a cardboard material sheet to form a unitary die cut body includes locating the first adhesive receiving area at the distal resealable closure flap segment and the adhesive portion at the proximal resealable closure flap segment; further comprising providing a release located at the resealable closure flap between the proximal resealable closure flap segment and the distal resealable closure flap segment; wherein stamping a cardboard material sheet to form a unitary die cut body includes stamping a cardboard material sheet to form a unitary die cut body with an inner panel extending upwardly from the bottom panel at the opening, wherein after opening the container with the pull-tab tear away strip, the removable covering is removeable from the adhesive portion with the adhesive portion adhering the proximal resealable closure flap segment to the inner panel, thereby placing the unitary die cut body container in the second closed configuration.

2. The method of claim 1, further comprising providing a padded insert assembly configured for insertion into the receiving cavity.

3. The method of claim 2, wherein providing a padded insert assembly configured for insertion into the receiving cavity includes providing a padded insert assembly with two horizontally extending pads which are affixed to and extend between two opposing walls, and wherein the padded insert assembly is configured for mechanized insertion into the receiving cavity.

4. A method for forming a round trip container, comprising: stamping a cardboard material sheet to form a unitary die cut body, wherein the unitary die cut body includes a top panel, a bottom panel opposite the top panel with the bottom panel having a first bottom panel end and a second bottom panel end, a first side panel, a second side panel, a resealable closure flap having a proximal resealable closure flap segment and a distal resealable closure flap segment, and an end panel, wherein a proximal end of the resealable closure flap is coupled to the top panel, whereby when the unitary die cut body is folded, the folded unitary die cut body defines a unitary die cut body container having a receiving cavity with an opening providing access to the receiving cavity, and wherein the resealable closure flap is configured to extend over the opening with a proximal end of the resealable closure flap operably coupled with the first end of the top panel of the unitary die cut body container; providing a tear-away strip between the proximal resealable closure flap segment and the distal resealable closure flap segment; placing a first adhesive receiving area at one of the resealable closure flap and the second end of the bottom panel of the unitary die cut body container that is opposite the first end of the unitary die cut body container, wherein the first adhesive receiving area is configured to receive adhesive to adhere the resealable closure flap to the unitary die cut body container when the unitary die cut body container is placed in a first closed configuration; placing an adhesive portion located on one of the resealable closure flap and the second end of the unitary die cut body container, wherein a removable covering is disposed over the adhesive portion when the unitary die cut body container is placed in the first closed configuration, and wherein the adhesive portion is configured to adhere the closure flap to the unitary die cut body container when the unitary die cut body container is placed in a second closed configuration; wherein stamping a cardboard material sheet to form a unitary die cut body includes locating the first adhesive receiving area at the distal resealable closure flap segment and the adhesive portion at the proximal resealable closure flap segment; providing a release located at the resealable closure flap between the proximal resealable closure flap segment and the distal resealable closure flap segment; wherein stamping a cardboard material sheet to form a unitary die cut body includes stamping a cardboard material sheet to form a unitary die cut body with an inner panel extending upwardly from the bottom panel at the opening, wherein after opening the

container with the pull-tab tear away strip, the removable covering is removeable from the adhesive portion with the adhesive portion adhering the proximal resealable closure flap segment to the inner panel, thereby placing the unitary die cut body container in the second closed configuration; providing a padded insert assembly configured for insertion into the receiving cavity; providing a first label portion on the unitary die cut body container having a face surface that includes first shipment information; providing a second label portion having a second face surface configured to have second shipment information printed thereon; and providing an adhesive located at the second label portion, wherein the adhesive releasably adheres the second label portion to the first label portion, and wherein the second label portion is configured to be removed from the first label portion and adhered to the container.

5. The method of claim 4, further comprising providing a transparent pouch configured to retain the first label portion and the second label portion to the unitary die cut body container.

6. The method of claim 5, wherein providing a transparent pouch includes providing a transparent pouch with a viewing window located in the pouch, and wherein the face surface is oriented to be visible through the viewing window.

7. The method of claim 6, wherein providing a transparent pouch includes providing a transparent pouch that extends over and covers the combined label.

8. The method of claim 7, wherein a lower surface of the first label portion includes a release backing, wherein the adhesive is located on a second lower surface of the second label portion, and wherein the face surface and the second face surface each face outwardly from the combined label on opposing sides thereof.
